Project Configuration Report

matter_wifi_soc_multi_sensor_app_demo_krasamo

Target Device:

Target Part: SIWG917M111MGTBA
Boards:
- SiWG917 Dev Kit Board (BRD2605A Rev A02)

Target SDK

Simplicity SDK Suite v2026.6.1: Bluetooth 11.1.1, Bluetooth Mesh SDK 11.0.1, Connect SDK 4.3.1.0, EmberZNet 9.1.1.0, Multiprotocol 1.1.1.0, OpenThread 3.1.1.0 (GitHub-fb274efe6), RAIL SDK 3.1.1.0, Silicon Labs Matter 2.9.1-1.6, Wi-Fi SDK 4.1.1-rc.2, Z-Wave SDK 8.1.1.0
SDK Version: 2026.6.1

Project Generators

• VS Code (GCC)

Installed Software Components:

• Board
- **Radio Board**
  BRD2605A
  Configurable: false

• Clusters
- **General**
  Access Control Server Cluster
  Sources: AccessControlEventHelper.cpp, CodegenIntegration.cpp, ArlEncoder.cpp, access-control-cluster.cpp
  Configurable: false

  Administrator Commissioning Server Cluster
  Sources: CodegenIntegration.cpp, AdministratorCommissioningLogic.cpp, AdministratorCommissioningCluster.cpp
  Configurable: false

  Basic Information Cluster
  Sources: CodegenIntegration.cpp
  Configurable: false

  Descriptor Cluster
  Sources: CodegenIntegration.cpp, DescriptorCluster.cpp
  Configurable: false

  Fixed Label Server Cluster
  Sources: CodegenIntegration.cpp, FixedLabelCluster.cpp

Configurable: false

General Commissioning Server Cluster
Sources: CodegenIntegration.cpp, GeneralCommissioningCluster.cpp
Configurable: false

General Diagnostics Server Cluster
Sources: CodegenIntegration.cpp, GeneralDiagnosticsCluster.cpp
Configurable: false

Group Key Management Server Cluster
Sources: CodegenIntegration.cpp, GroupKeyManagementCluster.cpp
Configurable: false

ICD Management Server Cluster
Sources: ICDManager.cpp, ICDMonitoringTable.cpp, CodegenIntegration.cpp, ICDConfigurationData.cpp, ICDManagementCluster.cpp, ICDNotifier.cpp
Configurable: false

Identify Server Cluster
Sources: IdentifyCluster.cpp, CodegenIntegration.cpp
Configurable: false

Localization Configuration Server Cluster
Sources: CodegenIntegration.cpp, LocalizationConfigurationCluster.cpp
Configurable: false

Network Commissioning Cluster
Sources: NetworkCommissioningCluster.cpp, ThreadScanResponse.cpp, WifiScanResponse.cpp, CodegenInstance.cpp
Configurable: false

OTA Software Update Requestor Cluster
Sources: ExtendedOTARequestorDriver.cpp, CodegenIntegration.cpp, OTARequestorAttributes.cpp, DefaultOTARequestor.cpp, BDXDownloader.cpp, OTARequestorCluster.cpp, DefaultOTARequestorStorage.cpp, DefaultOTARequestorDriver.cpp, OTATestEventTriggerHandler.cpp
Configurable: false

Operational Credentials Server Cluster
Sources: CodegenIntegration.cpp, OperationalCredentialsCluster.cpp
Configurable: false

Software Diagnostics Server Cluster
Sources: CodegenIntegration.cpp, SoftwareDiagnosticsCluster.cpp, software-fault-listener.cpp
Configurable: false

Time Format Localization Server Cluster
Sources: CodegenIntegration.cpp, TimeFormatLocalizationCluster.cpp
Configurable: false

User Label Server Cluster
Sources: CodegenIntegration.cpp, UserLabelCluster.cpp
Configurable: false

Wi-Fi Network Diagnostics Server Cluster
Sources: WiFiNetworkDiagnosticsCluster.cpp, CodegenIntegration.cpp
Configurable: false

- Measurement

Occupancy Sensing Server Cluster
Sources: CodegenIntegration.cpp, OccupancySensingCluster.cpp
Configurable: false

Relative Humidity Measurement Server Cluster
Sources: CodegenIntegration.cpp, RelativeHumidityMeasurementCluster.cpp
Configurable: false

Temperature Measurement Server Cluster
Sources: CodegenIntegration.cpp, TemperatureMeasurementCluster.cpp
Configurable: false

- Common

WiSeConnect Common Library
Sources: sl_utility.c
Configurable: false

- Device

- Si91x

- MCU

MCU Subsystem
Configurable: false

- Device Part
  - SIWG917

SIWG917M111MGTBA
Sources: system_si91x.c, startup_si91x.c
Configurable: false

- HAL
  Soft Reset
  Sources: sl_si91x_hal_soc_soft_reset.c
  Configurable: false

- Hardware
  Button
  Sources: sl_si91x_button.c
  Configurable: true
  Configs: sl_si91x_button_init_btn1_config.h, sl_si91x_button_init_btn0_config.h, sl_si91x_button_pin_config.h

  LED
  Sources: sl_si91x_rgb_led.c, sl_si91x_led.c
  Configurable: true
  Configs: sl_si91x_led_config.h, sl_si91x_rgb_led_init_led0_config.h, sl_si91x_led_init_led0_config.h, sl_si91x_rgb_led_config.h

- Peripheral
  I2C
  Configurable: true
  Configs: sl_si91x_i2c_init_i2c2_config.h

  ULP Timer
  Configurable: true
  Configs: sl_si91x_ulp_timer_init_timer0_config.h, sl_si91x_ulp_timer_common_config.h

- Service
  NVM3 Common Flash
  Sources: sl_si91x_nvm3_hal_flash.c, sl_si91x_common_flash_intf.c
  Configurable: false

- Memory Configuration
  MEMORY_MCU_ADVANCED_WIRELESS_BASIC
  Configurable: false

- Wireless
  BLE
  Sources: rsi_utils.c, sl_si91x_ble.c, rsi_common_apis.c, rsi_ble_gap_apis.c, rsi_bt_ble.c, rsi_bt_common_apis.c, rsi_ble_gatt_apis.c
  Configurable: false

  SiWx91x Wireless Subsystem
  Sources: sl_rsi_utility.c, rsi_hal_mcu_m4_rom.c, sli_si91x_driver.c, sl_net_si91x.c, sl_net_si91x_callback_framework.c, sl_si91x_bus.c, rsi_hal_mcu_m4_ram.c, sl_net_rsi_utility.c, sl_si91x_timer.c, sli_net_si91x_utility.c, efx32_ncp_host.c, sl_si91x_driver.c, sl_net_si91x_integration_handler.c, sli_siwx917_soc.c, sl_platform_wireless.c, siwx917_soc_ncp_host.c, sl_platform.c

Configurable: false

• Buffer Manager

Basic Buffer Manager

Sources: malloc_buffers.c

Configurable: false

• Matter-WiFi

Matter WiFi

Configurable: true

Configs: sl_matter_wifi_config.h

- Platform

Platform for SiWx917

Sources: SoftwareFaultReports.cpp, MatterConfig.cpp, BaseApplication.cpp, silabs_utils.cpp

Configurable: false

• Tinycrypt

Matter MbedTLS Integration

Sources: CHIPCryptoPALTinyCrypt.cpp, ecc.c, ecc_dh.c, ecc_dsa.c, tinycrypt_util.c

Configurable: false

- lwip

Matter LwIP

Sources: sys_arch.c

Configurable: false

• Matter

- Core

app_common

Sources: privilege-storage.cpp, util.cpp, DataModelHandler.cpp, attribute-storage.cpp, attribute-table.cpp, generic-callback-stubs.cpp, reporting.cpp

Configurable: false

• Config

Matter ZAP Driven DM

Configurable: false

- Examples

matter_multi_sensor_app

Configurable: false

- Platform

Configuration Over SWO
Configurable: false

Matter LED Support
Configurable: false

- CMSIS

CMSIS-RTOS2 Initialization
Sources: os_systick.c
Configurable: false

- Display

Matter Default LCD/Display Configuration
Configurable: false

- Low-power

ICD Server Configuration
Configurable: true
Configs: sl_matter_icd_config.h

- Security

Flash Attestation Certificate provisioning configuration
Sources: ProvisionStorageCustom.cpp, ProvisionStorageFlash.cpp
Configurable: true
Configs: sl_matter_provision_config.h

- Sensors

Si70xx Sensor Support
Sources: Si70xxSensorGsdk.cpp, Si70xxSensorWiseMcu.cpp
Configurable: false

- Utilities

Assert Functions
Sources: sl_assert.c
Configurable: false

Single Link List
Sources: sl_slist.c
Configurable: false

- Utils

Logging to UART
Configurable: false

- Protocol

  Wi-Fi

  Sources: sl_wifi_callback_framework.c, sl_wifi.c, sli_wifi_callback_framework.c

  Configurable: false

- RTOS

  - FreeRTOS

  FreeRTOS Kernel

  Sources: event_groups.c, timers.c, list.c, queue.c, stream_buffer.c, tasks.c, croutine.c

  Configurable: true

  Configs: FreeRTOSConfig.h

- Resources

  WiSeConnect Resources

  Configurable: true

  Configs: sl_net_default_values.h, lwipopts.h, sl_wifi_region_db_config.h, aws_iot_config.h

- Service

  - Network Management

  - Configuration

    Basic Network Configuration Manager

    Sources: sli_net_credentials.c, sl_net_basic_profiles.c, sl_net_credentials.c

    Configurable: false

- Stack

  Matter Core Components

  Configurable: true

  Configs: SilabsTracingConfig.h, sl_matter_config.h, ICDServerBuildConfig.h, build_config.h

  OTA Requestor

  Sources: OTAConfig.cpp, OTAImageProcessorImpl.cpp

  Configurable: true

  Configs: sl_matter_ota_config.h

  - Bluetooth

  Bluetooth Network Layer

  Configurable: false

  - Shell

  Matter Shell

  Sources: Engine.cpp, cmd_otcli.cpp, Config.cpp, WiFi.cpp, streamer.cpp, MainLoopSilabs.cpp, EventHandlerLibShell.cpp, ICDShellCommands.cpp, Ota.cpp, TracingShellCommands.cpp, IMShellCommands.cpp, BLE.cpp, Help.cpp, streamer_silabs.cpp, CommandSet.cpp, MatterShell.cpp, OnboardingCodes.cpp, BLEShellCommands.cpp, ShellCommands.cpp, globals.cpp, Meta.cpp, Dns.cpp, Stat.cpp, cmd_misc.cpp, Device.cpp

  Configurable: false

- Third Party
  - **Segger**
    - RTT

      SEGGER RTT

      Sources: SEGGER_RTT.c

      Configurable: true

      Configs: SEGGER_RTT_Conf.h

Pins:

Peripherals:

Peripheral	Software Component	Custom Peripheral Name
I2C0		
I2C1		
ULP_I2C	I2C (i2c2)	
USART0		
UART1		
QEI		
OPAMP1		
OPAMP2		
OPAMP3		
ULP_UART		
SSI_MASTER		
SSI_SLAVE		
HSPI_SECONDARY		
ULP_SSI		
GSPI_MASTER		
I2S0		
ULP_I2S		
SCT		
PWM_CH0		
PWM_CH1		
PWM_CH2		
PWM_CH3		
PWM		
SDC_CH1		
SDC_CH2		
SDC_CH3		
SDC_CH4		
ADC_CH1		
ADC_CH2		
ADC_CH3		
ADC_CH4		
ADC_CH5		
ADC_CH6		

ADC_CH7		
ADC_CH8		
ADC_CH9		
ADC_CH10		
ADC_CH11		
ADC_CH12		
ADC_CH13		
ADC_CH14		
ADC_CH15		
ADC_CH16		
COMP1		
COMP2		
DAC0		
DAC1		
SYSRTC		
UULP_VBAT_GPIO		
SDIO		
GPIO		
MCU_CLK_OUT		